CURRICULUM PROPOSAL

Data Science & AI Engineering

Highly Applied Focus

An intensive, project-driven 135-hour roadmap built around real-world industrial implementation.

45
LIVE SESSIONS
Hands-on interactive learning

135
TOTAL HOURS
3 Sessions/wk (3 hrs each)

12+
PHASE PROJECTS
Real portfolio deliverables

60%
PRACTICAL CODE
Live coding in every session

PRACTICAL-FIRST METHODOLOGY

Built Around Continuous Live Coding & Real Projects

60% Live Code & Lab Sprints
Every single session contains active coding

Students don't passively watch slides. Every session includes a 60-minute live coding sprint where learners write, test, and debug code alongside instructor guidance.

12 Phase Milestone Projects
One full project per curriculum phase

Learners solidify concepts by building modular projects—from SQL analytical pipelines to fine-tuned LLM agents and containerized PyTorch microservices.

Industry Benchmark Datasets
Zero toy datasets used

All labs utilize noisy, messy industrial data sourced from fintech, e-commerce, healthcare, and enterprise logs to simulate true workplace challenges.

Deployed Production Capstone
Live API + Dashboard + Cloud

Students deploy an end-to-end AI product using Docker, FastAPI, and PowerBI, presented in front of an expert industry evaluation panel.

SESSIONS 01 – 07 (21 HOURS)

Phase 1 & 2: Python, Git & Math for AI

Phases 01 & 02

SESSION	CORE TOPIC	HANDS-ON LAB / APPLICATION	LIVE PROJECT BUILT
SESS 01	Python & Environment Setup	Configuring Anaconda, VS Code, primitive types & structures.	✓ Python CLI Tool
SESS 02	Control Flow & Functional Code	Building custom data parsers with list comprehensions & lambdas.	✓ Automated Log Parser
SESS 03	OOP & Professional Git Workflow	Class inheritance, GitHub branching, pull requests & conflict resolution.	✓ Open Source PR Lab
SESS 04	Linear Algebra: Vectors & Matrices	Matrix transformations, dot products, vector spaces in Python.	✓ Vector Engine Lab
SESS 05	Multivariate Calculus & Optimization	Computing Jacobians, loss functions & gradient descent from scratch.	✓ Custom Optimizer
SESS 06	Probability & Random Variables	Bayes theorem simulations & continuous/discrete probability modeling.	✓ Risk Simulator
SESS 07	Inferential Statistics & A/B Testing	Hypothesis testing, Z-scores, t-tests, ANOVA on real user data.	✓ A/B Test Evaluator

SESSIONS 08 – 14 (21 HOURS)

Phase 3 & 4: Data Manipulation & Visualization

Phases 03 & 04

SESSION	CORE TOPIC	HANDS-ON LAB / APPLICATION	LIVE PROJECT BUILT
SESS 08	NumPy High-Performance Arrays	Vectorized operations, multi-dimensional array manipulation & indexing.	✓ Matrix Computing Engine
SESS 09	Pandas Cleaning & Wrangling	Cleaning missing data, string sanitization, and type transformations.	✓ Messy Data Pipeline
SESS 10	Advanced Pandas Aggregation	Groupby, pivot tables, multi-index operations, dataset joins.	✓ E-Commerce ETL Pipeline
SESS 11	Time-Series Data Wrangling	Resampling, rolling windows, date-time indexing, lag features.	✓ Financial Series Parser
SESS 12	Matplotlib Publication Visuals	Custom publication-grade line plots, scatter charts, and subplots.	✓ Custom Chart Theme
SESS 13	Seaborn Statistical Graphics	Heatmaps, pair plots, box plots, statistical distribution checks.	✓ Correlation Heatmap Suite
SESS 14	Interactive Plotly Dashboards	Building dynamic maps, hoverable charts & interactive web visual tools.	✓ Dynamic Web Dashboard

SESSIONS 15 – 21 (21 HOURS)

Phase 5 & 6: SQL Analytics & Exploratory Data Analysis

Phases 05 & 06

SESSION	CORE TOPIC	HANDS-ON LAB / APPLICATION	LIVE PROJECT BUILT
SESS 15	SQL Foundations & Filtering	Writing optimized SELECT queries, filtering, aggregation & GROUP BY.	✓ Retail Database Queries
SESS 16	Complex Joins & Subqueries	INNER/LEFT/RIGHT joins, CTEs, and nested subquery optimization.	✓ Multi-Table Analytics
SESS 17	Window Functions & Partitions	ROW_NUMBER, RANK, LEAD, LAG, and running totals over partitions.	✓ User Cohort Analyzer
SESS 18	Python-SQL Data Pipelines	Connecting PostgreSQL to Pandas DataFrames via SQLAlchemy.	✓ SQL-to-Pandas Bridge
SESS 19	Systematic Univariate EDA	Outlier detection, distribution skewness correction, data scaling.	✓ Data Quality Audit
SESS 20	Automated Profiling Tools	Accelerating EDA using YData Profiling & Sweetviz modules.	✓ Auto EDA Generator
SESS 21	Comprehensive EDA Case Study	End-to-end investigation & feature discovery on raw industry data.	✓ Churn EDA Project

SESSIONS 22 – 27 (18 HOURS)

Phase 7: Core Machine Learning Algorithms

Phase 07

SESSION	CORE TOPIC	HANDS-ON LAB / APPLICATION	LIVE PROJECT BUILT
SESS 22	Linear & Polynomial Regression	OLS model fitting, residual analysis, MSE/R2 valuation.	✓ Real Estate Valuation
SESS 23	Regularization (L1 Lasso / L2 Ridge)	Preventing overfitting, feature selection via ElasticNet shrinkage.	✓ Regularized Predictor
SESS 24	Logistic Regression & Classification	Sigmoid decision boundaries, ROC-AUC curves, confusion matrices.	✓ Credit Default Classifier
SESS 25	kNN & Naive Bayes Classifiers	Distance metrics, probabilistic classification on text/features.	✓ Spam Classification Engine
SESS 26	Support Vector Machines (SVM)	Hyperplane margin maximization & non-linear kernel tricks.	✓ Medical Diagnosis Model
SESS 27	Unsupervised Clustering Methods	K-Means, Hierarchical clustering, and DBSCAN density grouping.	✓ Customer Segmentation

SESSIONS 28 – 32 (15 HOURS)

Phase 8: Advanced Ensembles & Optimization

Phase 08

SESSION	CORE TOPIC	HANDS-ON LAB / APPLICATION	LIVE PROJECT BUILT
SESS 28	Decision Trees & Random Forests	Gini impurity splitting, bagging ensembles, feature importance.	✓ Employee Attrition Model
SESS 29	Gradient Boosting (XGBoost / LightGBM)	Boosting residual optimization, CatBoost handling categorical data.	✓ XGBoost Fraud Engine
SESS 30	Dimensionality Reduction	Principal Component Analysis (PCA), t-SNE & UMAP projection.	✓ High-Dim Feature Reducer
SESS 31	Bayesian Hyperparameter Tuning	GridSearch, RandomSearch & Optuna automated search spaces.	✓ Automated Optuna Pipeline
SESS 32	Imbalanced Datasets (SMOTE)	Oversampling, undersampling, focal loss, precision-recall tuning.	✓ Fraud Imbalance Model

SESSIONS 33 – 36 (12 HOURS)

Phase 9: Deep Learning & Computer Vision

Phase 09

SESSION	CORE TOPIC	HANDS-ON LAB / APPLICATION	LIVE PROJECT BUILT
SESS 33	PyTorch Neural Architecture	Tensors, Autograd engine, custom nn.Module training loops.	✓ PyTorch Perceptron
SESS 34	Neural Optimization & Regularization	AdamW optimizers, Dropout, Batch Normalization, LR schedulers.	✓ Deep Net Optimizer
SESS 35	Convolutional Networks & ResNet	Image convolutions, pooling, Transfer Learning with ImageNet models.	✓ Vision Image Classifier
SESS 36	Sequence Models (LSTMs & GRUs)	Recurrent networks for sequential time-series & text processing.	✓ Stock Predictor LSTM

SESSIONS 37 – 40 (12 HOURS)

Phase 10: Generative AI, LLMs & RAG Systems

Phase 10

SESSION	CORE TOPIC	HANDS-ON LAB / APPLICATION	LIVE PROJECT BUILT
SESS 37	Transformers & Self-Attention	Multi-head attention mechanisms, tokenization with HuggingFace.	✓ Attention Visualizer
SESS 38	Vector DBs & Embeddings	ChromaDB, FAISS, Pinecone setup & high-dimensional vector search.	✓ Semantic Search Engine
SESS 39	LangChain & Enterprise RAG	Document parsing, chunking, retriever pipelines over private PDF files.	✓ PDF Knowledge Assistant
SESS 40	LLM Fine-Tuning (PEFT & LoRA)	Parameter-efficient fine-tuning on domain datasets via HuggingFace.	✓ Domain-Tuned LLM

SESSIONS 41 – 45 (15 HOURS)

Phase 11 & 12: MLOps, Business Intelligence & Capstone

Phases 11 & 12

SESSION	CORE TOPIC	HANDS-ON LAB / APPLICATION	LIVE PROJECT BUILT
SESS 41	FastAPI Inference Microservice	Exposing machine learning models as production REST API endpoints.	✓ Live Predictor REST API
SESS 42	Docker Containerization & MLflow	Packaging models into Docker containers & tracking metrics with MLflow.	✓ Dockerized ML Service
SESS 43	Power BI Enterprise Dashboards	Connecting database sources to Power BI, writing DAX measures.	✓ Executive BI Dashboard
SESS 44	Executive Business Storytelling	Translating model ROC-AUC scores into financial ROI metrics.	✓ Business ROI Deck
SESS 45	Deployed Capstone Presentation	Live defense of end-to-end deployed AI cloud system to panel.	✓ Final Production Capstone

PEDAGOGICAL STRUCTURE

Anatomy of Every 3-Hour Applied Session

3-Hour Session Blueprint

45 Mins: Code Review & Debugging
Instructor-led breakdown of student lab submissions and error logs.

75 Mins: Theory & Architectural Deep Dive
Mathematical mechanics, algorithm trade-offs, and system design.

60 Mins: Live Coding Sprint
Hands-on exercise building real pipeline components in real time.

Grading & Practical Assessment

Weekly Practical Labs (12 Labs)	35%
Phase Milestone Projects	25%
Deployed Production Capstone	30%
Code Review & GitHub Activity	10%

TRACK PROPOSAL SUMMARY

A World-Class, Project-Driven Data Science & AI Curriculum

Transforming students into job-ready Data Scientists & AI Engineers through 45 live sessions and 12+ real-world projects.

45 Sessions
135 Total Live Hours

12+ Live Projects
Full Portfolio Deliverables

60% Practical Ratio
Live Coding In Every Class